Books

Godmann, H. R., Haaf, J. M., **Visser**, I., Szymanik, J., & Sarafoglou, A. (2026). *Toward a Principled Bayesian Workflow in Semantics and Pragmatics: A Tutorial*. Cambridge University Press. https://doi.org/10.31219/osf.io/wck5s_v2

Visser, I., & Speekenbrink, M. (2022). *Mixture and Hidden Markov Models with R*. Springer Nature. https://doi.org/10.1007/978-3-031-01440-6

Work in Progress

18. Ijdo, N. K., Eckhardt, C. L., **Visser**, I., Twisk, J., Cnossen, M. H., Fijnvandraat, K., Oosterlaan, J., & Königs, M. (2026, **under review**). Exploring Neurocognitive Functioning in Toddlers with Sickle Cell Disease with Eye-Tracking. *Anemia*.
17. Johnson, T., Sarafoglou, A., Haaf, J. M., **Visser**, I., & Szymanik, J. (2026, **under review**). Modeling Scalar Categorization: Comparing Threshold-Based and Prototype-Based Representations. https://doi.org/10.31234/osf.io/vd6bh_v1
16. Kolber, N., Röseler, L., Back, M., Evans, T., Hartmann, H., Korbmacher, M., Le Foll, E., Müller, M., Rahal, R.-M., Richter, M., Seibold, H., Silverstein, P., Steltenpohl, C. N., Syed, M., Vaidis, D., **Visser**, I., Verheyen, S., Wallrich, L., & Azevedo, F. (2026, **under review**). Creating a Journal in the Age of Open Science: The Case of Replication Research (R2). *LIBER Quarterly*. https://osf.io/preprints/metaarxiv/zctw7_v1
15. Maat, S. C., Dreuning, K. M., Janssen, S. J., **Visser**, I., Hunnius, S., Stevens, M. F., de Graaff, J. C., van Heurn, L. E., Königs, M., Oosterlaan, J., & Derikx, J. P. (2026, **under review**). General Anesthesia Exposure and Neurocognitive Outcome in Infants.
14. Pfadt, J., Bartos, F., Godmann, H. R., Wagenmakers, E.-J., Sarafoglou, A., **Visser**, I., Pawel, S., Aust, F., de Ron, J., Wiechert, S., Siepe, B., Sekulovski, N., Marsman, M., van den Bergh, D., & Donzallaz, M. (2026, **under review**). The Prevalence of Bayesian Inference, Preregistration, and Open Data in Psychology, 2004--2024. *Nature Human Behaviour*. https://doi.org/10.31234/osf.io/ck3js_v3
13. Ploeger, A., Karami, A., & **Visser**, I. (2026, **under review**). An Evolutionary Perspective on Depressive Symptoms: A Network Analysis. *Evolutionary Behavioral Sciences*.
12. Schmidt, D., **Visser**, I., Maurits, L., & Thiele, M. (2026, **under review**). Eye-Tracking Data Quality Across Human Infants, Adults, and Chimpanzees: Challenges and Solutions for Developmental and Comparative Psychology. *Behavior Research Methods*. https://osf.io/preprints/psyarxiv/cptgu_v1
11. **Visser**, I., Geambașu, A., Kachergis, G., Oliveira, C. M., Rocha-Hidalgo, J., Zettersten, M., Ahamat, T. M., Alessandroni, N., Althaus, N., Arunachalam, S., Aussems, S., Axelsson, E., Aydin, Z., Baumgartner, H. A., Bergmann, C., Bettoni, R., Black, A., Bulf, H., Byers-Heinlein, K., … Levelt, C. (2026, **under review**). ManyBabies 3: A Multi-Lab Study of Infant Algebraic Rule Learning. https://doi.org/10.31234/osf.io/ghrdt_v2
10. Zaharieva, M., Collonesi, C., & **Visser**, I. (2026, **in preparation**). Endogenous Attention, Emotion Regulation, Feeding, and Sleep in 3- to 13-Month-Old Infants.
9. Sarafoglou, A., Giacobello, A., Godmann, H., Johnson, T., **Visser**, I., Haaf, J. M., & Szymanik, J. (2025, **under review**). Are Semantic Representations Stable? A Bayesian Framework. https://doi.org/10.31219/osf.io/hvgb7_v2
8. Schuwerk, T., Kampis, D., Alessandroni, N., Altvater-Mackensen, N., Arias-Trejo, N., Axelsson, E. L., Baillargeon, R., Baumann, A.-E., Bernard, C., Biro, S., Blankenship, T. L., Blomberg, I., Bohn, M., Bradford, E. E. F., Byers-Heinlein, K., Canudas Grabolosa, I., Chen, E. M., Chen, X., Corbit, J., … Rakoczy, H. (2025, **under review**). Action Anticipation Based on an Agent's Epistemic State in Toddlers and Adults. https://osf.io/preprints/psyarxiv/x4jbm_v4
7. van Kordelaar, J., **Visser**, I., van Vliet, E., & van Weijen, D. (2025, **under review**). Rubric Use in Writing Assessment: Relations Between Assessors' Experience and Satisfaction.
6. Wassink-de Stigter, R., Jansen, R., Deen, M., **Visser**, I., Simons, I., & Soenen, S. (2025, **under review**). Subtypes of Families with Multiple and Complex Problems Within Integrated Specialized Family-Focused Mental Health Care.
5. Zaharieva, M., Collonesi, C., & **Visser**, I. (2025, **in preparation**). Growing Emotion and Attention Regulation: a Toolbox for Studying Self-Regulation in Infancy.

4. Kosie, J., Zettersten, M., Abu-Zhaya, R., Amso, D., Babineau, M., Baumgartne, H., Belia, M., Benavides, S., Bergmann, C., Berteletti, I., et al. (2023, **under review**). ManyBabies 5: A Large-Scale Investigation of the Proposed Shift from Familiarity Preference to Novelty Preference in Infant Looking Time. https://doi.org/10.31234/osf.io/ck3vd
3. Kucharský, Š., Raijmakers, M., & **Visser**, I. (2023, **in preparation**). Strategy Classification in Economic Games Using Eye-Movement Data.
2. Zaharieva, M., Kucharský, Š., & **Visser**, I. (2023, **under review**). Habituation, Part I. Design Choices in the Infant Habituation Paradigm: A Pre-Registered Crowd-Sourced Systematic Review and Meta-Analysis.
1. Hofman, A., Kievit, R., Stevenson, C., Molenaar, D., **Visser**, I., & van der Maas, H. (2018, **under review**). *The Dynamics of the Development of Mathematics Skills: A Comparison of Theories of Developing Intelligence.*

Publications

105. Alessandroni, N., Miller, R., Altschul, D., Barrett, L. P., Bazhydai, M., Elsherif, M., Espinosa, J., Gjoneska, B., Héjja-Brichard, Y., Mazza, V., Paukner, A., Pronizius, E., Proulx, M., Qadri, M., Reilly, O., Schwing, R., Sebastián-Enesco, C., Šlipogor, V., de Sousa, A., **Visser**, I., et al. (2026). Flexible Behavior or Flexible Methods? A Cross-Taxon Review of Experimental Designs in Reversal Learning. *Open Mind: Discoveries in Cognitive Science*. https://osf.io/mvche
104. Korthals, L., Akrong, E., Geller, G., Rosenbusch, H., Grasman, R., & **Visser**, I. (2026). Towards Reliable LLM Grading Through Self-Consistency and Selective Human Review: Higher Accuracy, Less Work. *Machine Learning and Knowledge Extraction*, *8*(3), 74. https://doi.org/10.3390/make8030074
103. McMillan, B., Baumgartner, H., Bergmann, C., Frank, M. C., Hamlin, J. K., Kampis, D., Kline Struhl, M., Ko, E.-S., Kosie, J., Lew-Williams, C., Lucca, K., Schuwerk, T., Soderstrom, M., **Visser**, I., Yuen, F., Zettersten, M., & Byers-Heinlein, K. (2026). What 5000 Babies Can Tell Us About Developing Minds and How to Study Them. *Communications Psychology*.
102. Vaidis, D. C., Miranda, J. F., Buchanan, E. M., Yang, Y. F., Kowal, M., Schmidt, K., Topor, M., Misiak, M., Miller, R., Protzko, J., Gjoneska, B., Miller, J. K., Exner, A., Azevedo, F., Paruzel-Czachura, M., Mushtaq, F., Oliveira, C., Wagge, J. R., De Moor, D., ... Pronizius, E. (2026). The Advantage of Big Team Science: Lessons Learned from Cognitive Science. *Collabra: Psychology*, *12*(1), 160129. https://doi.org/10.1525/collabra.160129
101. Alessandroni, N., Altschul, D., Baumgartner, H., Bazhydai, M., Brosnan, S., Byers-Heinlein, K., Call, J., Chittka, L., Elsherif, M., Espinosa, J., Freeman, M., Gjoneska, B., Güntürkün, O., Huber, L., Krasheninnikova, A., Mazza, V., Miller, R., Moreau, D., Nawroth, C., ... Prétôt, L. (2025). Challenges and Promises of Big Team Comparative Cognition. *Nature Human Behaviour*, *9*(2), 240–242. https://doi.org/10.1038/s41562-024-02081-6
100. Korthals, L., Rosenbusch, H., Grasman, R., & **Visser**, I. (2025). Grading University Students with LLMs: Performance and Acceptance of a Canvas-Based Automation. In *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium, Blue Sky, and WideAIED* (pp. 36–43). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-99264-3_5
99. Lucca, K., Capelier-Mourguy, A., Cirelli, L., Byers-Heinlein, K., Dal Ben, R., Frank, M. C., Henderson, A. M., Kominsky, J. F., Liberman, Z., Margoni, F., et al. (2025). Infants' Social Evaluation of Helpers and Hinderers: A Large-Scale, Multi-Lab, Coordinated Replication Study. *Developmental Science*, *28*(1). https://doi.org/10.1111/desc.13581
98. Speekenbrink, M., & **Visser**, I. (2025). State-Dependent Missingness in Hidden Markov Models, with an Application to Drop-Out in a Clinical Trial. *Psychometrika*, *90*(2), 476–507. https://doi.org/10.1017/psy.2024.3
97. **Visser**, I. (2025). Hidden Markov Models. In M. C. Frank & A. Majid (Eds.), *Open Encyclopedia of Cognitive Science*. MIT Press. https://doi.org/10.21428/e2759450.528dddc3
96. Alessandroni, N., Altschul, D., Bazhydai, M., Byers-Heinlein, K., Elsherif, M., Gjoneska, B., Huber, L., Mazza, V., Miller, R., Nawroth, C., Pronizius, E., Qadri, M., Šlipogor, V., Soderstrom, M., Stevens, J., **Visser**, I., Williams, M., Zettersten, M., & Prétôt, L. (2024). Comparative Cognition Needs Big Team Science: How Large-Scale Collaborations Will Unlock the Future of the Field. *Comparative Cognition and Behavior Reviews*, *19*, 67–72. https://doi.org/10.3819/CCBR.2024.190001

95. Exner, A., Bettoni, R., Cantiani, C., Koleini, A., Oliveira, C. M., Thompson, A., **Visser**, I., & Zettersten, M. (2024). The Relationship Between Rule Learning in Infancy (5-12 Months) and Language Skills at 24-30 Months.
94. Hoogeveen, S., Borsboom, D., Kucharský, Š., Marsman, M., Molenaar, D., de Ron, J., Sekulovski, N., **Visser**, I., van Elk, M., & Wagenmakers, E.-J. (2024). Prevalence, Patterns and Predictors of Paranormal Beliefs in The Netherlands: a Several-Analysts Approach. *Royal Society Open Science*, *11*(9), 240049. https://doi.org/10.1098/rsos.240049
93. Johnson, T., Sarafoglou, A., Haaf, J. M., **Visser**, I., & Szymanik, J. (2024). Comparing the Threshold and Prototype Model for Gradable Adjectives. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *46*.
92. Kucharský, Š., Zaharieva, M., Raijmakers, M., & **Visser**, I. (2024). Habituation, Part II. Rethinking the Habituation Paradigm. *Infant and Child Development*, *33*(1), e2383. https://doi.org/10.1002/icd.2383
91. Schaaf, J. V., Johansson, A., **Visser**, I., & Huizenga, H. M. (2024). What's in a Name: The Role of Verbalization in Reinforcement Learning. *Psychonomic Bulletin and Review*, *31*(6), 2746–2757. https://doi.org/10.3758/s13423-024-02506-3
90. Zaharieva, M. S., Salvadori, E. A., Messinger, D. S., **Visser**, I., & Colonnesi, C. (2024). Automated Facial Expression Measurement in a Longitudinal Sample of 4-And 8-Month-Olds: Baby FaceReader 9 and Manual Coding of Affective Expressions. *Behavior Research Methods*, *56*(6), 5709–5731. https://doi.org/10.3758/s13428-023-02301-3
89. Exner, A., Mayor, J., Moreau, D., Oliveira, C. M., Soderstrom, M., Tauzin, T., **Visser**, I., Weyers, I., & Zettersten, M. (2023). Assessing Test-Retest Reliability of Rule Learning Measures in Infants. https://doi.org/10.17605/OSF.IO/MA8Z5
88. Frank, M., Kunda, N., Lavechin, M., Oudeyer, P.-Y., Saxe, R., de Seyssel, M., & **Visser**, I. (2023). 4.3 Group 2.1: Embodied Intention Prediction Challenge. In J. M. Rehg, P.-Y. Oudeyer, L. B. Smith, S. Tsuji, S. Stojanov, & N. A. Thai (Eds.), *Developmental Machine Learning: From Human Learning to Machines and Back (Dagstuhl Seminar 22422)* (p. 162). https://doi.org/10.4230/DagRep.12.10.143
87. Franse, R., van Schijndel, T. J., **Visser**, I., & Raijmakers, M. (2023). Children's Understanding of Floating and Sinking: Predictions and Explanations Tell Different Stories.
86. Geambașu, A., Spit, S., van Renswoude, D., Blom, E., Fikkert, P. J., Hunnius, S., Junge, C. C., Verhagen, J., **Visser**, I., Wijnen, F., & Levelt, C. C. (2023). Robustness of the Rule-Learning Effect in 7-Month-Old Infants: A Close, Multicenter Replication of Marcus Et Al.(1999). *Developmental Science*, *26*(1), e13244. https://doi.org/10.1111/desc.13244
85. Lichtenberg, L., **Visser**, I., & Raijmakers, M. (2023). Latent Markov Models to Test the Strategy Use of 3-Year-Olds in a Rule-Based Feedback-Learning Task. *Multivariate Behavioral Research*, *59*(6), 1123–1136. https://doi.org/10.1080/00273171.2023.2170963
84. Schaaf, J., Huizenga, H., **Visser**, I., & Jepma, M. (2023). Context Valence Affects Learning Mechanisms and Confidence, but not Learning, in Adolescents.
83. Spit, S., Geambașu, A., Renswoude, D. v., Blom, E., Fikkert, P., Hunnius, S., Junge, C., Verhagen, J., **Visser**, I., Wijnen, F., & Levelt, C. C. (2023). Robustness of the Cognitive Gains in 7-Month-Old Bilingual Infants: A Close Multi-Center Replication of Kovács and Mehler (2009). *Developmental Science*, *26*(6), e13377. https://doi.org/10.1111/desc.13377
82. **Visser**, I., Kucharský, Š., Levelt, C., Stefan, A. M., Wagenmakers, E.-J., & Oakes, L. (2023). Bayesian Sample Size Planning for Developmental Studies. *Infant and Child Development*, *33*(1), e2412. https://doi.org/10.1002/icd.2412
81. Wesarg-Menzel, C., Ebbes, R., Hensums, M., Wagemaker, E., Zaharieva, M., Staaks, J., van den Akker, A., **Visser**, I., Hoeve, M., Brummelman, E., Dekkers, T., Schuitema, J., Larsen, H., Colonnesi, C., Jansen, B., Overbeek, G., Huizenga, H., & Wiers, R. (2023). Development and Socialization of Self-Regulation from Infancy to Adolescence: A Meta-Review Differentiating Between Self-Regulatory Abilities, Goals, and Motivation. *Developmental Review*, *69*, 101090. https://doi.org/https://doi.org/10.1016/j.dr.2023.101090
80. Jepma, M., Schaaf, J. V., **Visser**, I., & Huizenga, H. M. (2022). Impaired Learning to Dissociate Advantageous and Disadvantageous Risky Choices in Adolescents. *Scientific Reports*, *12*(1), 1–14. https://doi.org/10.1038/s41598-022-10100-7
79. Lüken, M., Kucharský, Š., & **Visser**, I. (2022). Characterising Eye Movement Events with an Unsupervised Hidden Markov Model. *Journal of Eye Movement Research*, *15*(1). https://doi.org/10.16910/jemr.15.1.4

78. Schaaf, J. V., Xu, B., Jepma, M., **Visser**, I., & Huizenga, H. M. (2022). (Mal) Adaptive Learning After Switches Between Object-Based and Rule-Based Environments. *Computational Brain & Behavior*, *5*(2), 157–167. https://doi.org/10.1007/s42113-022-00134-5
77. **Visser**, I. (2022). No Data Left Behind. *Infant and Child Development*, *31*(5), e2339. https://doi.org/10.1002/icd.2339
76. **Visser**, I., Bergmann, C., Byers-Heinlein, K., Dal Ben, R., Duch, W., Forbes, S., Franchin, L., Frank, M., Geraci, A., Hamlin, J. K., et al. (2022). Improving the Generalizability of Infant Psychological Research: The ManyBabies Model. *Behavioral and Brain Sciences*, *45*. https://doi.org/10.1017/s0140525x21000455
75. **Visser**, I., & Speekenbrink, M. (2022a). Hidden Markov Models. In *Mixture and Hidden Markov Models with R* (pp. 125–172). Springer International Publishing. https://doi.org/10.1007/978-3-031-01440-6_4
74. Byers-Heinlein, K., Tsui, R., van Renswoude, D., Black, A., Barr, R., Brown, A., Colomer, M., Durrant, S., Gonzalez-Gomez, N., Hay, J., Hernik, M., Jartó, M., Kovács, Á., Laoun-Rubenstein, A., Lew-Williams, C., Liszkowski, U., Liu, L., Noble, C., Potter, C., ... Singh, L. (2021). The Development of Gaze Following in Monolingual and Bilingual Infants: A Multi-Laboratory Study. *Infancy*, *26*(1), 4–38. https://doi.org/10.1111/infa.12360
73. Jepma, M., Schaaf, J. V., **Visser**, I., & Huizenga, H. M. (2021). Effects of Advice on Experienced-Based Learning in Adolescents and Adults. *Journal of Experimental Child Psychology*, *211*, 105230. https://doi.org/10.1016/j.jecp.2021.105230
72. Kucharský, Š., Tran, N.-H., Veldkamp, K., Raijmakers, M., & **Visser**, I. (2021). Hidden Markov Models of Evidence Accumulation in Speeded Decision Tasks. *Computational Brain and Behavior*, *4*(4), 416–441. https://doi.org/10.1007/s42113-021-00115-0
71. Kucharský, Š., van Renswoude, D., Raijmakers, M., & **Visser**, I. (2021). WALD-EM: Wald Accumulation for Locations and Durations of Eye Movements. *Psychological Review*, *128*(4), 667. https://doi.org/10.1037/rev0000292
70. Kuijpers, R. E., **Visser**, I., & Molenaar, D. (2021). Testing the Within-State Distribution in Mixture Models for Responses and Response Times. *Journal of Educational and Behavioral Statistics*, *46*(3), 348–373. https://doi.org/10.3102/1076998620957240
69. **Visser**, I., & Speekenbrink, M. (2021). *Package 'Hmmr'*. https://doi.org/10.32614/cran.package.hmmr
68. Jepma, M., Schaaf, J. V., **Visser**, I., & Huizenga, H. M. (2020). Uncertainty-Driven Regulation of Learning and Exploration in Adolescents: A Computational Account. *PLoS Computational Biology*, *16*(9), e1008276. https://doi.org/10.1371/journal.pcbi.1008276
67. Kucharský, Š., Houtkoop, B. L., & **Visser**, I. (2020). Code Sharing in Psychological Methods and Statistics: An Overview and Associations with Conventional and Alternative Research Metrics.
66. Kucharský, Š., **Visser**, I., Trutescu, G.-O., Laurence, P. G., Zaharieva, M., & Raijmakers, M. E. (2020). Cognitive Strategies Revealed by Clustering Eye Movement Transitions. *Journal of Eye Movement Research*, *13*(1). https://doi.org/10.16910/jemr.13.1.1
65. Lüken, M., Kucharský, Š., & **Visser**, I. (2020). *Preregistration: Estimating the Parameter Recovery of the Hidden Markov Model in GazeHMM*.
64. Mulder, K., Klugkist, I., van Renswoude, D., & **Visser**, I. (2020). Mixtures of Peaked Power Batschelet Distributions for Circular Data with Application to Saccade Directions. *Journal of Mathematical Psychology*, *95*, 102309. https://doi.org/10.1016/j.jmp.2019.102309
63. van Bers, B. M., van Schijndel, T. J., **Visser**, I., & Raijmakers, M. E. (2020). Cognitive Flexibility Training Has Direct and Near Transfer Effects, but No Far Transfer Effects, in Preschoolers. *Journal of Experimental Child Psychology*, *193*, 104809. https://doi.org/10.1016/j.jecp.2020.104809
62. van Renswoude, D. R., Raijmakers, M. E., & **Visser**, I. (2020). Looking (for) Patterns: Similarities and Differences Between Infant and Adult Free Scene-Viewing Patterns. *Journal of Eye Movement Research*, *13*(1). https://doi.org/10.16910/jemr.13.1.2
61. Bayarri, M., Berger, J. O., Jang, W., Ray, S., Pericchi, L. R., & **Visser**, I. (2019). Prior-Based Bayesian Information Criterion. *Statistical Theory and Related Fields*, *3*(1), 2–13. https://doi.org/10.1080/24754269.2019.1582126
60. Berger, J., Jang, W., Ray, S., Rericchi, L. R., & **Visser**, I. (2019). Rejoinder by James Berger, Woncheol Jang, Surajit Ray, Luis R. Pericchi and Ingmar Visser. *Statistical Theory and Related Fields*, *3*(1), 37–39. https://doi.org/10.1080/24754269.2019.1611147

59. Schaaf, J. V., Jepma, M., **Visser**, I., & Huizenga, H. M. (2019). A Hierarchical Bayesian Approach to Assess Learning and Guessing Strategies in Reinforcement Learning. *Journal of Mathematical Psychology*, *93*, 102276. https://doi.org/10.1016/j.jmp.2019.102276
58. van Renswoude, D. R., van den Berg, L., Raijmakers, M. E., & **Visser**, I. (2019). Infants' Center Bias in Free Viewing of Real-World Scenes. *Vision Research*, *154*, 44–53. https://doi.org/10.1016/j.visres.2018.10.003
57. van Renswoude, D. R., **Visser**, I., Raijmakers, M. E., Tsang, T., & Johnson, S. P. (2019). Real-World Scene Perception in Infants: What Factors Guide Attention Allocation? *Infancy*, *24*(5), 693–717. https://doi.org/10.1111/infa.12308
56. Dutilh, G., Annis, J., Brown, S., Cassey, P., Evans, N., Grasman, R., Hawkins, G., Heathcote, A., Holmes, W., Krypotos, A.-M., Kupitz, C., Leite, F., Lerche, V., Lin, Y.-S., Logan, G., Palmeri, T., Starns, J., Trueblood, J., van Maanen, L., … Donkin, C. (2018). The Quality of Response Time Data Inference: A Blinded, Collaborative Assessment of the Validity of Cognitive Models. *Psychonomic Bulletin & Review*, *26*(4), 1051–1069. https://doi.org/10.3758/s13423-017-1417-2
55. Hofman, A. D., **Visser**, I., Jansen, B. R., Marsman, M., & van der Maas, H. L. (2018). Fast and Slow Strategies in Multiplication. *Learning and Individual Differences*, *68*, 30–40. https://doi.org/10.1016/j.lindif.2018.09.007
54. van Renswoude, D. R., Raijmakers, M. E., Koornneef, A., Johnson, S. P., Hunnius, S., & **Visser**, I. (2018). Gazepath: An Eye-Tracking Analysis Tool that Accounts for Individual Differences and Data Quality. *Behavior Research Methods*, *50*(2), 834–852. https://doi.org/10.3758/s13428-017-0909-3
53. van Schijndel, T. J., Huijpen, K., **Visser**, I., & Raijmakers, M. E. (2018). Investigating the Development of Causal Inference by Studying Variability in 2-To 5-Year-Olds' Behavior. *PLoS ONE*, *13*(4), e0195019. https://doi.org/10.1371/journal.pone.0195019
52. Molenaar, D., **Visser**, I., et al. (2017). Cognitive and Psychometric Modelling of Responses and Response Times. *British Journal of Mathematical & Statistical Psychology*, *70*(2), 185–186. https://doi.org/10.1111/bmsp.12102
51. **Visser**, I., & Poessé, R. (2017). Parameter Recovery, Bias and Standard Errors in the Linear Ballistic Accumulator Model. *British Journal of Mathematical and Statistical Psychology*, *70*(2), 280–296. https://doi.org/10.1111/bmsp.12100
50. **Visser**, I., Schmittmann, V. D., & Raijmakers, M. E. (2017). Markov Process Models for Discrimination Learning. In *Longitudinal Models in the Behavioral and Related Sciences* (pp. 337–366). Routledge. https://doi.org/10.4324/9781315091655-14
49. Blakey, E., **Visser**, I., & Carroll, D. J. (2016). Different Executive Functions Support Different Kinds of Cognitive Flexibility: Evidence from 2-, 3-, and 4-Year-Olds. *Child Development*, *87*(2), 513–526. https://doi.org/10.1111/cdev.12468
48. Jansen, B. R., Hofman, A. D., Savi, A., **Visser**, I., & van der Maas, H. L. (2016). Self-Adapting the Success Rate When Practicing Math. *Learning and Individual Differences*, *51*, 1–10. https://doi.org/10.1016/j.lindif.2016.08.027
47. Souverein, F. A., Ward, C. L., **Visser**, I., & Burton, P. (2016). Serious, Violent Young Offenders in South Africa: Are They Life-Course Persistent Offenders? *Journal of Interpersonal Violence*, *31*(10), 1859–1882. https://doi.org/10.1177/0886260515570748
46. van Renswoude, D., Johnson, S., Raijmakers, M., & **Visser**, I. (2016). Do Infants Have the Horizontal Bias? *Infant Behavior and Development*, *44*, 38–48. https://doi.org/10.1016/j.infbeh.2016.05.005
45. Hofman, A. D., **Visser**, I., Jansen, B. R., & van der Maas, H. L. (2015). The Balance-Scale Task Revisited: A Comparison of Statistical Models for Rule-Based and Information-Integration Theories of Proportional Reasoning. *PLoS One*, *10*(10), e0136449. https://doi.org/10.1371/journal.pone.0136449
44. Hosenfeld, B., Bos, E. H., Wardenaar, K. J., Conradi, H. J., van der Maas, H. L., **Visser**, I., & de Jonge, P. (2015). Major Depressive Disorder as a Nonlinear Dynamic System: Bimodality in the Frequency Distribution of Depressive Symptoms Over Time. *Bmc psychiatry*, *15*(1), 1–9. https://doi.org/10.1186/s12888-015-0596-5
43. van Schijndel, T. J., **Visser**, I., van Bers, B. M., & Raijmakers, M. E. (2015). Preschoolers Perform More Informative Experiments After Observing Theory-Violating Evidence. *Journal of experimental child psychology*, *131*, 104–119. https://doi.org/10.1016/j.jecp.2014.11.008
42. Andersen, L. M., **Visser**, I., Crone, E. A., Koolschijn, P., & Raijmakers, M. E. (2014). Cognitive Strategy Use as an Index of Developmental Differences in Neural Responses to Feedback. *Developmental psychology*, *50*(12), 2686. https://doi.org/10.1037/a0038106

41. Raijmakers, M. E., Schmittmann, V. D., & **Visser**, I. (2014). Costs and Benefits of Automatization in Category Learning of Ill-Defined Rules. *Cognitive psychology*, *69*, 1–24. https://doi.org/10.1016/j.cogpsych.2013.12.002
40. van Bers, B. M., **Visser**, I., & Raijmakers, M. (2014). Preschoolers Learn to Switch with Causally Related Feedback. *Journal of Experimental Child Psychology*, *126*, 91–102. https://doi.org/10.1016/j.jecp.2014.03.007
39. Van Bers, B., **Visser**, I., & Raijmakers, M. (2014). The Distinctive Effects of Exogenous Factors on Preschoolers' DCCS Performance. *Manuscript submitted for publication*.
38. **Visser**, I. (2014). Why Reproducible Reporting?
37. **Visser**, I., & Speekenbrink, M. (2014a). Comments on: Latent Markov Models: a Review of a General Framework for the Analysis of Longitudinal Data with Covariates. *Test*, *23*, 478–483. https://doi.org/10.1007/s11749-014-0389-z
36. **Visser**, I., & Speekenbrink, M. (2014b). It's a Catastrophe! Testing Dynamics Between Competing Cognitive States Using Mixture and Hidden Markov Models. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.), *CogSci 2014* (pp. 1688–1693, Vol. 2). Cognitive Science Society.
35. Ihrke, M., Behrendt, J., Schrobsdorff, H., **Visser**, I., & Hasselhorn, M. (2013). Negative Priming Persists in the Absence of Response-Retrieval. *Experimental Psychology*, *60*(1), 12–21. https://doi.org/10.1027/1618-3169/a000169
34. Nivard, M., van der Sluis, S., Franic, S., Cramer, A., **Visser**, I., Middeldorp, C., Lubke, G., Boomsma, D., & Borsboom, D. (2012). Phenotypic Complexity as Genetic Dark Matter: A Network Explanation of Missing Heritability. *Proceedings of the 42nd Annual Meeting of the Behavior-Genetics-Association*.
33. **Visser**, I., & Raijmakers, M. E. (2012). Developing Representations of Compound Stimuli. *Frontiers in psychology*, *3*, 73. https://doi.org/10.3389/fpsyg.2012.00073
32. Dutilh, G., Wagenmakers, E.-J., **Visser**, I., & van der Maas, H. L. (2011). A Phase Transition Model for the Speed-Accuracy Trade-Off in Response Time Experiments. *Cognitive Science*, *35*(2), 211–250. https://doi.org/10.1111/j.1551-6709.2010.01147.x
31. Huizenga, H. M., **Visser**, I., & Dolan, C. V. (2011). Testing Overall and Moderator Effects in Random Effects Meta-Regression. *British Journal of Mathematical and Statistical Psychology*, *64*(1), 1–19. https://doi.org/10.1348/000711010x522687
30. van Bers, B. M., **Visser**, I., van Schijndel, T. J., Mandell, D. J., & Raijmakers, M. E. (2011). The Dynamics of Development on the Dimensional Change Card Sorting Task. *Developmental Science*, *14*(5), 960–971. https://doi.org/10.1111/j.1467-7687.2011.01045.x
29. **Visser**, I. (2011a). Methodological Solipsism. In P. Hogan (Ed.), *The Cambridge Encyclopedia of Language Sciences* (p. 497). Cambridge University Press.
28. **Visser**, I. (2011b). Seven Things to Remember About Hidden Markov Models: A Tutorial on Markovian Models for Time Series. *Journal of Mathematical Psychology*, *55*(6), 403–415. https://doi.org/10.1016/j.jmp.2011.08.002
27. Pronk, T., & **Visser**, I. (2010). The Role of Reversal Frequency in Learning Noisy Second Order Conditional Sequences. *Consciousness and cognition*, *19*(2), 627–635. https://doi.org/10.1016/j.concog.2009.12.008
26. van Duijvenvoorde, A. C., Jansen, B. R., **Visser**, I., & Huizenga, H. M. (2010). Affective and Cognitive Decision-Making in Adolescents. *Developmental neuropsychology*, *35*(5), 539–554. https://doi.org/10.1080/87565641.2010.494749
25. **Visser**, I. (2010). Review of: W. Zucchini, IL MacDonald (2009) Hidden Markov Models for Time Series: an Introduction Using R. *Journal of Mathematical Psychology*, *54*(6), 509–511. https://doi.org/10.1016/j.jmp.2010.07.001
24. **Visser**, I., Jansen, B. R., & Speekenbrink, M. (2010). A Framework for Discrete Change. In P. Molenaar & K. Newell (Eds.), *Individual Pathways of Change: Statistical Models for Analyzing Learning and Development* (pp. 109–123). APA Science. https://doi.org/10.1037/12140-007
23. **Visser**, I., & Speekenbrink, M. (2010). DepmixS4: An R Package for Hidden Markov Models. *Journal of Statistical Software*, *36*(7), 1–21.
22. **Visser**, I., Raijmakers, M. E., & Pothos, E. M. (2009). Individual Strategies in Artificial Grammar Learning. *The American journal of psychology*, *122*(3), 293–307. https://doi.org/10.2307/27784404
21. **Visser**, I., Raijmakers, M. E., & van der Maas, H. L. (2009). Hidden Markov Models for Individual Time Series. *Dynamic process methodology in the social and developmental sciences*, 269–289. https://doi.org/10.1007/978-0-387-95922-1_13

20. Borsboom, D., & **Visser**, I. (2008). Semantic Cognition or Data Mining? *Behavioral and Brain Sciences*, *31*(6), 714–715. https://doi.org/10.1017/s0140525x08005906
19. **Visser**, I., et al. (2008). Review of: J. Rissanen (2007) Information and Complexity in Statistical Modeling. *Kwantitatieve Methoden*, *2007*, 1–2.
18. Jansen, B. R., Raijmakers, M. E., & **Visser**, I. (2007). Rule Transition on the Balance Scale Task: a Case Study in Belief Change. *Synthese*, *155*, 211–236. https://doi.org/10.1007/s11229-006-9142-9
17. **Visser**, I. (2007). Depmix: An R-Package for Fitting Mixture Models on Mixed Multivariate Data with Markov Dependencies. *R-package manual*, *39*, 65. https://doi.org/10.32614/cran.package.depmix
16. **Visser**, I., Raijmakers, M. E., & Molenaar, P. C. (2007). Characterizing Sequence Knowledge Using Online Measures and Hidden Markov Models. *Memory & Cognition*, *35*(6), 1502–1517. https://doi.org/10.3758/bf03193619
15. Blaazer, T., & **Visser**, I. (2006). Estimating Correlations and Reliabilities of Implicit and Explicit Tests Using a Latent Variable Approach. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *28*(28).
14. Jansen, B. R., Raijmakers, M. E., & **Visser**, I. (2006). Learning on the Balance Scale Task. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *28*(28).
13. Schmittmann, V. D., **Visser**, I., & Raijmakers, M. E. (2006). Multiple Learning Modes in the Development of Performance on a Rule-Based Category-Learning Task. *Neuropsychologia*, *44*(11), 2079–2091. https://doi.org/10.1016/j.neuropsychologia.2005.12.011
12. **Visser**, I., Blaazer, T., et al. (2006). Estimating Correlations and Reliabilities of Implicit and Explicit Tests Using a Latent Variable Approach. *Proceedings of the Annual Meeting of the Cognitive Science Society*.
11. Huizinga, M., **Visser**, I., Hamaker, E., & van der Molen, M. (2005). The Development of Executive Control from Childhood Through Young-Adulthood: A Latent Variables Approach. In B. Hommel, G. Band, W. La Heij, & G. Wolters (Eds.), *Fourteenth Conference of the European Society for Cognitive Psychology: Proceedings* (p. 38). Leiden University.
10. Van Der Maas, H. L., Raijmakers, M. E., & **Visser**, I. (2005). Inferring the Structure of Latent Class Models Using a Genetic Algorithm. *Behavior research methods*, *37*, 340–352. https://doi.org/10.3758/bf03192702
9. **Visser**, I., Tagaro, J., & Huizinga, H. (2005). A Meta-Analysis of the Effects of the Secondary Tasks and Stimulus Complexity on Implicit Learning. In B. Hommel, G. Band, W. La Heij, & G. Wolters (Eds.), *Fourteenth Conference of the European Society for Cognitive Psychology: Proceedings*. Leiden University.
8. **Visser**, I. (2002). *Rules and Associations: Hidden Markov Models and Neural Networks in the Psychology of Learning* [Doctoral dissertation, Universiteit van Amsterdam].
7. **Visser**, I., Raijmakers, M. E., & Molenaar, P. (2002). Fitting Hidden Markov Models to Psychological Data. *Scientific Programming*, *10*(3), 185–199. https://doi.org/10.1155/2002/874560
6. **Visser**, I., Raijmakers, M. E., & Molenaar, P. C. (2001). Hidden Markov Model Model Interpretations of Neural Networks. *Connectionist Models of Learning, Development and Evolution: Proceedings of the Sixth Neural Computation and Psychology Workshop, Liège, Belgium, 16-18 September 2000*, 197. https://doi.org/10.1007/978-1-4471-0281-6_20
5. **Visser**, I., Raijmakers, M., & Molenaar, P. (2000). Reaction Times and Predictions in Sequence Learning: A Comparison. *Twenty-second annual conference of the Cognitive Science Society*.
4. **Visser**, I. (2000). Hidden Markov Model Interpretations of Neural Networks. *Behavioral and Brain Sciences*, *23*(4), 494–495. https://doi.org/10.1017/s0140525x00513351
3. **Visser**, I., Raijmakers, M. E., & Molenaar, P. (2000). Confidence Intervals for Hidden Markov Model Parameters. *British journal of mathematical and statistical psychology*, *53*(2), 317–327. https://doi.org/10.1348/000711000159240
2. **Visser**, I., Raijmakers, M., Molenaar, P., et al. (1998). Statistical Properties of Hidden Markov Models. *International Workshop on Advanced Black-Box Techniques for Nonlinear Modeling*.
1. **Visser**, I. (1996). *Mind Rules: a Philosophical Essay on Psychological Rules and the Rules of Psychology* (tech. rep. No. X–96–01). Amsterdam: Institute for Logic, Language and Computation.